

Water Cycle in a Bag

A Fun Science Experiment — Complete Lesson Plan

Grade Level	Subject	Duration	Format
Class 3 – 5	Science / EVS	45–60 Minutes	Hands-on Experiment

1. LESSON OVERVIEW

Wonder Question

Riya noticed puddles on the road after it rained.
The next morning — they were GONE!
Where did all the water disappear to?

Learning Objectives

By the end of this lesson students will be able to:

- ✓ Explain the three key stages of the water cycle (evaporation, condensation, precipitation).
- ✓ Connect each stage to real-life observations from daily life in India.
- ✓ Conduct a simple experiment and record observations like a scientist.
- ✓ Use correct vocabulary: evaporation, condensation, precipitation, water vapour.

2. KEY SCIENCE CONCEPTS

Evaporation ■■

When water is heated by the sun it turns into invisible water vapour and rises into the air.

Real-life example: Puddles disappearing on a sunny day, wet clothes drying on a clothesline.

Condensation ■

Water vapour rises and meets cooler air high up.
It turns back into tiny liquid droplets forming clouds.

Real-life example: Drops on the outside of a cold water bottle; morning dew on leaves.

Precipitation ■

When cloud droplets become heavy enough they fall back to Earth as rain, hail, or snow.

Real-life example: Kerala monsoon rains, Delhi drizzle, snow in Kashmir.

The Water Cycle — Round and Round!

Water is NEVER created or destroyed — it just keeps changing form.

Sun heats water → Evaporation (up as vapour) → Condensation (clouds form)

→ Precipitation (rain falls) → Water collects in rivers & oceans → Cycle repeats

3. MATERIALS NEEDED (per student / group)

#	Item	Purpose
1	Zip-lock / resealable plastic bag	The 'mini atmosphere'
2	Water (approx. ½ cup)	Represents oceans / lakes
3	Blue food colouring (a few drops)	Makes water visible
4	Permanent marker / sketch pen	Drawing on the bag
5	Clear adhesive tape	Sticking bag to window
6	A sunny window or lamp	Heat source — acts as the Sun

4. EXPERIMENT STEPS

Step 1 — Decorate Your Bag

Using a permanent marker, draw on the outside of the bag:

- A bright SUN at the top
- Wavy water lines at the bottom
- A fluffy CLOUD in the middle
- An UP arrow labelled 'Evaporation'
- A DOWN arrow labelled 'Rain / Precipitation'

Step 2 — Add Water

Pour about $\frac{1}{2}$ cup of water into the bag.
Add 2–3 drops of blue food colouring and mix gently.
The blue water represents the ocean or a lake.

Step 3 — Seal the Bag

Close the zip-lock seal firmly.
Run your fingers along the seal to make sure NO air leaks out.
Check twice — a leaky bag will not work!

Step 4 — Tape to a Sunny Window

Use clear tape on all four edges to stick the bag to a sunny window or near a lamp.
Make sure the bag is upright with the water at the bottom.

Step 5 — Observe & Record

Wait 30–60 minutes and observe every 10 minutes.
Look for: fog / mist inside → drops on upper surface → drips falling.
Draw what you see in your Science Journal.

5. STUDENT OBSERVATION TABLE

Students fill in this table as the experiment progresses:

Time	What I See Inside the Bag	Water Cycle Stage
0 min (start)		
10 min		
20 min		
30 min		
45 min		
60 min		

6. DISCUSSION QUESTIONS

During the experiment

- Q1. What did you first notice inside the bag after 10 minutes?
 Q2. Where did the drops form — top, bottom, or sides? Why?
 Q3. When the drops fell back down, which stage does that show?

Connecting to real life

- Q4. How is what happened in the bag similar to real rain?
 Q5. Why do we get more rain during monsoon season?
 Q6. Why do puddles disappear faster on a hot sunny day?

Think deeper

- Q7. What would happen if you put the bag in a dark cupboard?
 Q8. Is the water cycle ever-ending? Will it stop one day?
 Q9. How does the water cycle help farmers in your village/city?

7. WATER CYCLE ALL AROUND INDIA

Region	Water Cycle Observation
Kerala	Heavy monsoon rains June–September — intense precipitation
Rajasthan	Puddles dry very quickly — rapid evaporation in desert heat
Delhi / North India	Morning dew on grass and car windscreens — condensation
Kashmir	Snow in winter — precipitation in solid form
Tamil Nadu / Madurai	Hot, humid weather speeds up evaporation all year round

8. ASSESSMENT & EVALUATION

Assessment Type	What to Look For	Marks
Observation Table	Accuracy & detail of recorded observations	10
Diagram (in journal)	Correctly labelled water cycle diagram	10
Discussion answers	Understanding of evaporation, condensation, precipitation	10

Class chant / quiz	Ability to recall and use vocabulary	10
Take-home activity	Show water cycle bag to family & write 3 lines	10
	TOTAL	50

9. CLASS CHANT ■

EVAPORATION — UP!
CONDENSATION — COOL!
PRECIPITATION — DOWN!
WATER CYCLE — ALL AROUND!

10. TAKE-HOME ACTIVITY

Show Your Science to the Family!

1. Take your water cycle bag home — keep it sealed!
2. Tape it to a sunny window at home and show your family.
3. Explain: Evaporation → Condensation → Precipitation.
4. In your journal write 3 sentences about what you observed.
5. Draw a labelled water cycle diagram and colour it neatly.

Teacher Tips

- Set up the bags first thing in the morning so results are visible by mid-class.
- If sunlight is weak, place a desk lamp 10–15 cm from the bag.
- Encourage students to use Tamil / Hindi vocabulary too (Varsha, Baashpa).
- For extension: compare bags in sunlight vs shade — great inquiry question!

This lesson aligns with NCERT EVS / Science curriculum for Classes 3–5 · Prepared for classroom use in Indian schools